# Decision-Tree for Classification

MACHINE LEARNING WITH TREE-BASED MODELS IN PYTHON

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Data Scientist

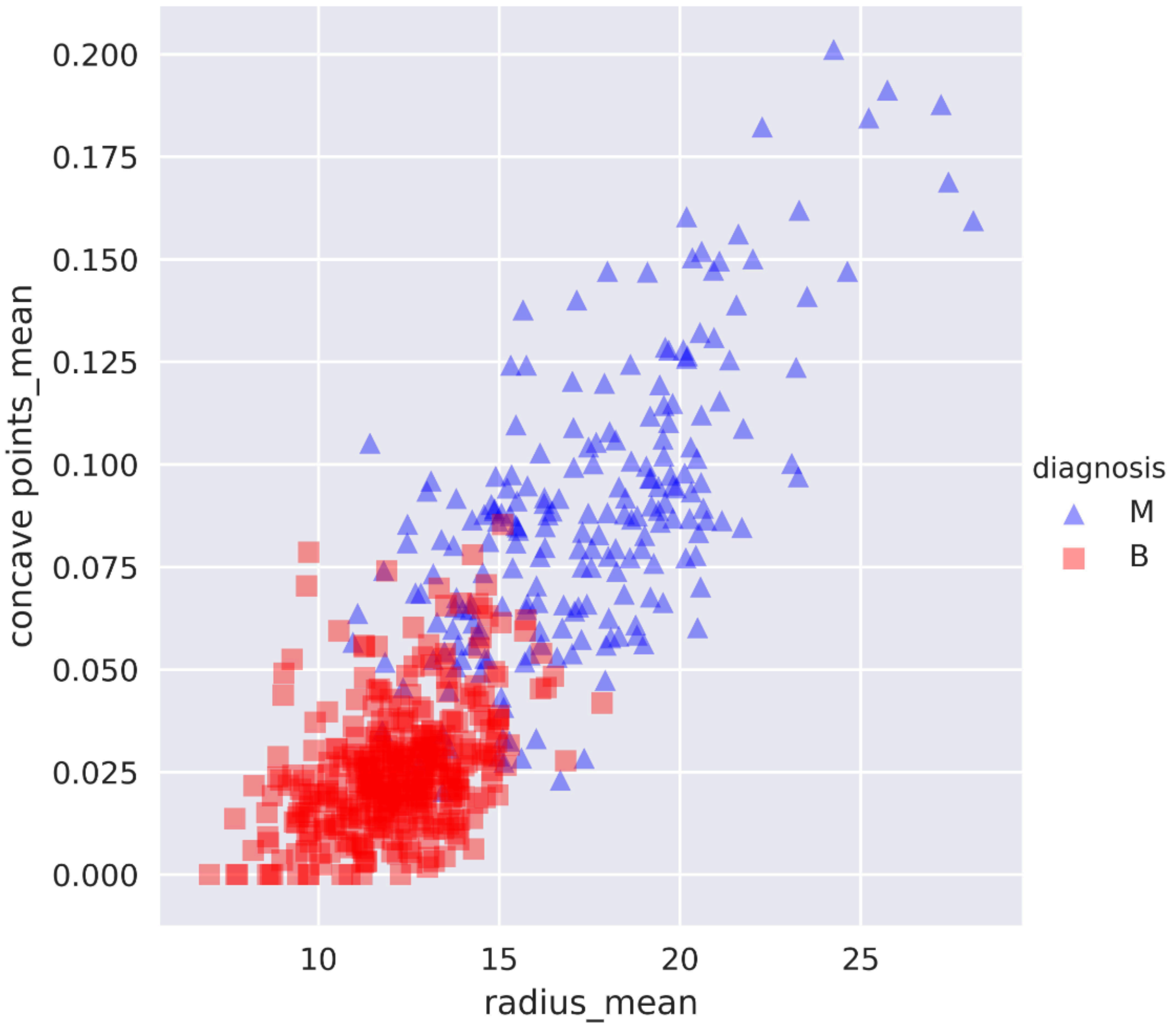
# Course Overview

- **Chap 1:** Classification And Regression Tree (CART)
- **Chap 2:** The Bias-Variance Tradeoff
- **Chap 3:** Bagging and Random Forests
- **Chap 4:** Boosting
- **Chap 5:** Model Tuning

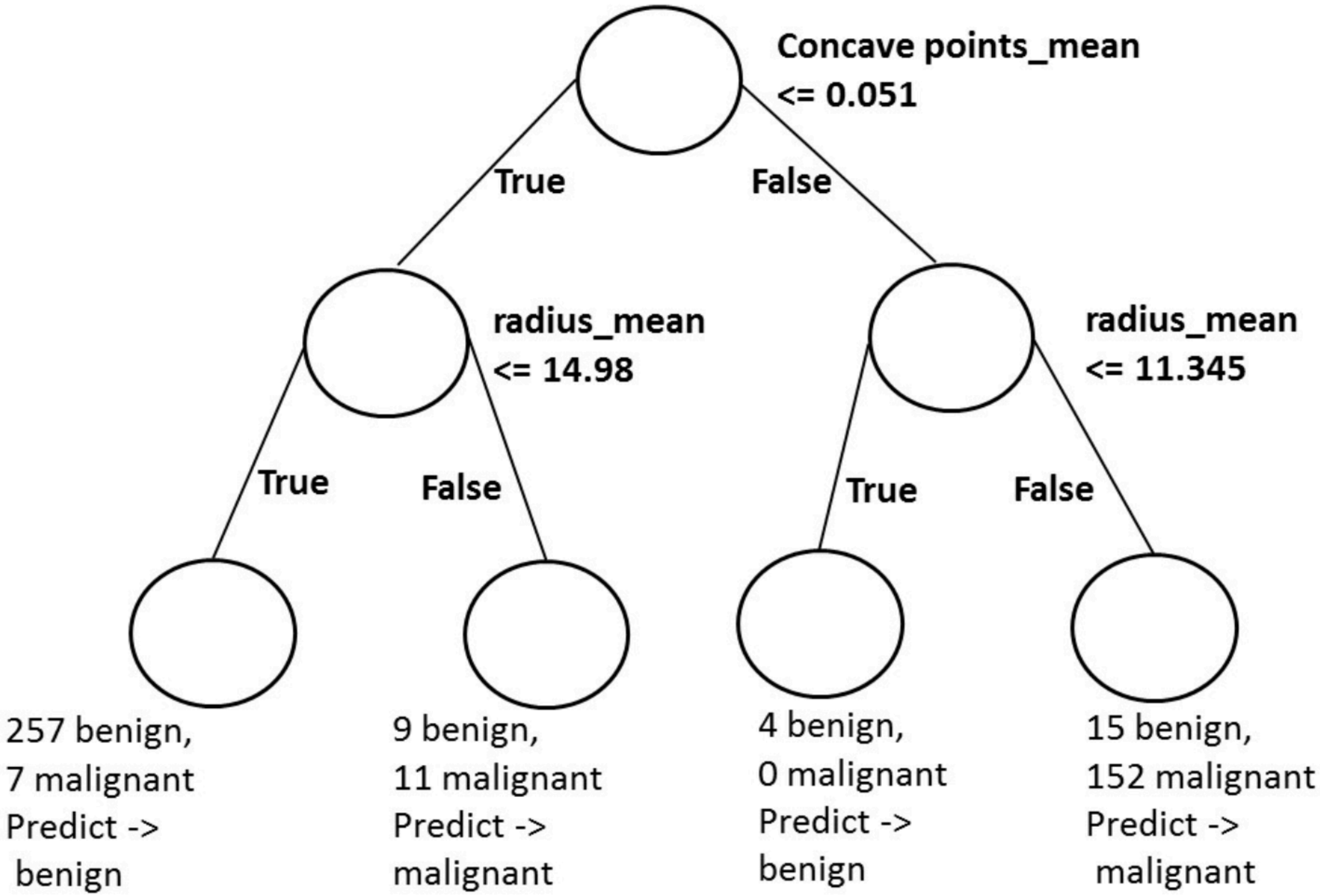
# Classification-tree

- Sequence of if-else questions about individual features.
- **Objective**: infer class labels.
- Able to capture non-linear relationships between features and labels.
- Don't require feature scaling (ex: Standardization, ..)

# Breast Cancer Dataset in 2D



# Decision-tree Diagram



# Classification-tree in scikit-learn

```
# Import DecisionTreeClassifier
from sklearn.tree import DecisionTreeClassifier
# Import train_test_split
from sklearn.model_selection import train_test_split
# Import accuracy_score
from sklearn.metrics import accuracy_score
# Split the dataset into 80% train, 20% test
X_train, X_test, y_train, y_test= train_test_split(X, y,
                                                    test_size=0.2,
                                                    stratify=y,
                                                    random_state=1)
# Instantiate dt
dt = DecisionTreeClassifier(max_depth=2, random_state=1)
```

# Classification-tree in scikit-learn

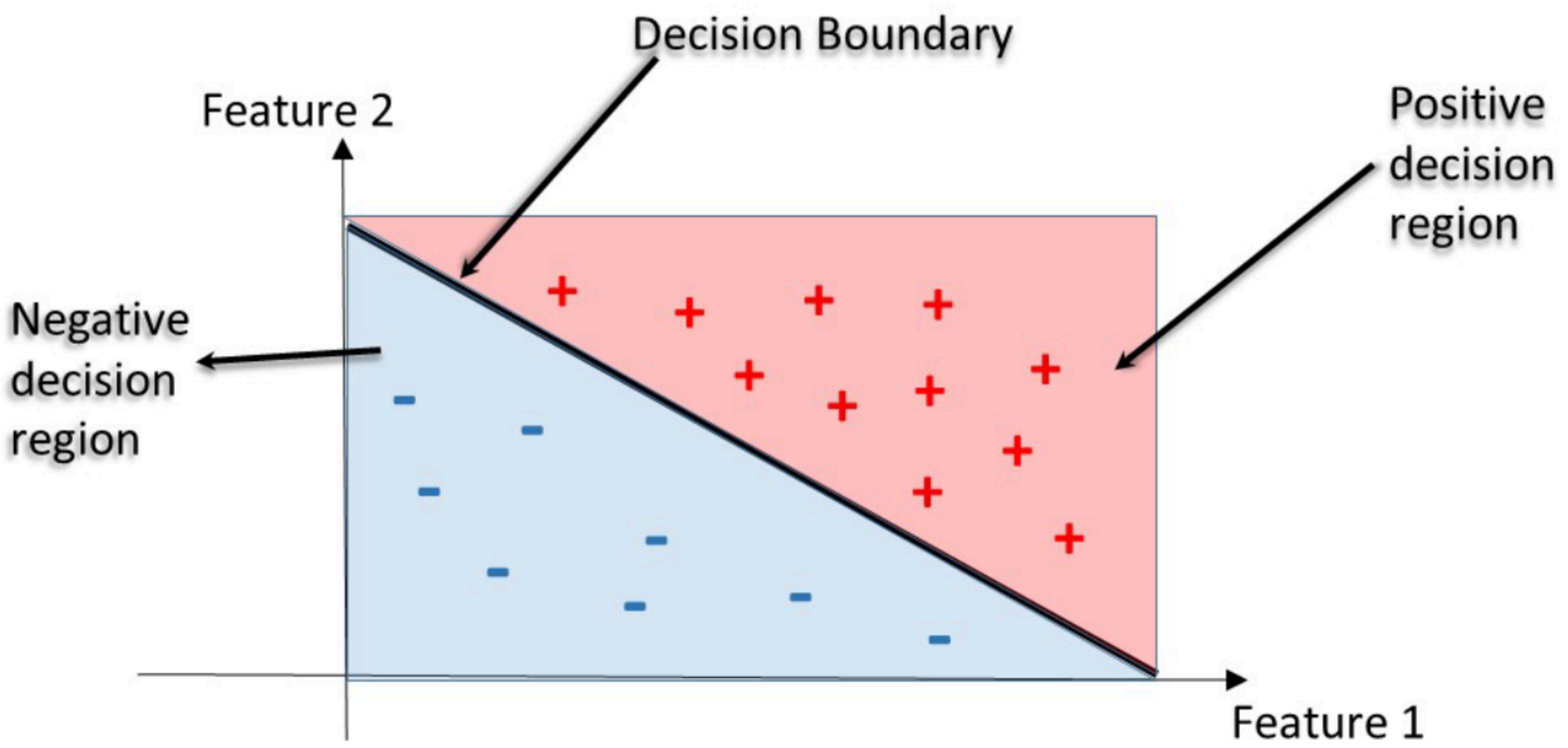
```
# Fit dt to the training set
dt.fit(X_train,y_train)

# Predict the test set labels
y_pred = dt.predict(X_test)
# Evaluate the test-set accuracy
accuracy_score(y_test, y_pred)
```

```
0.90350877192982459
```

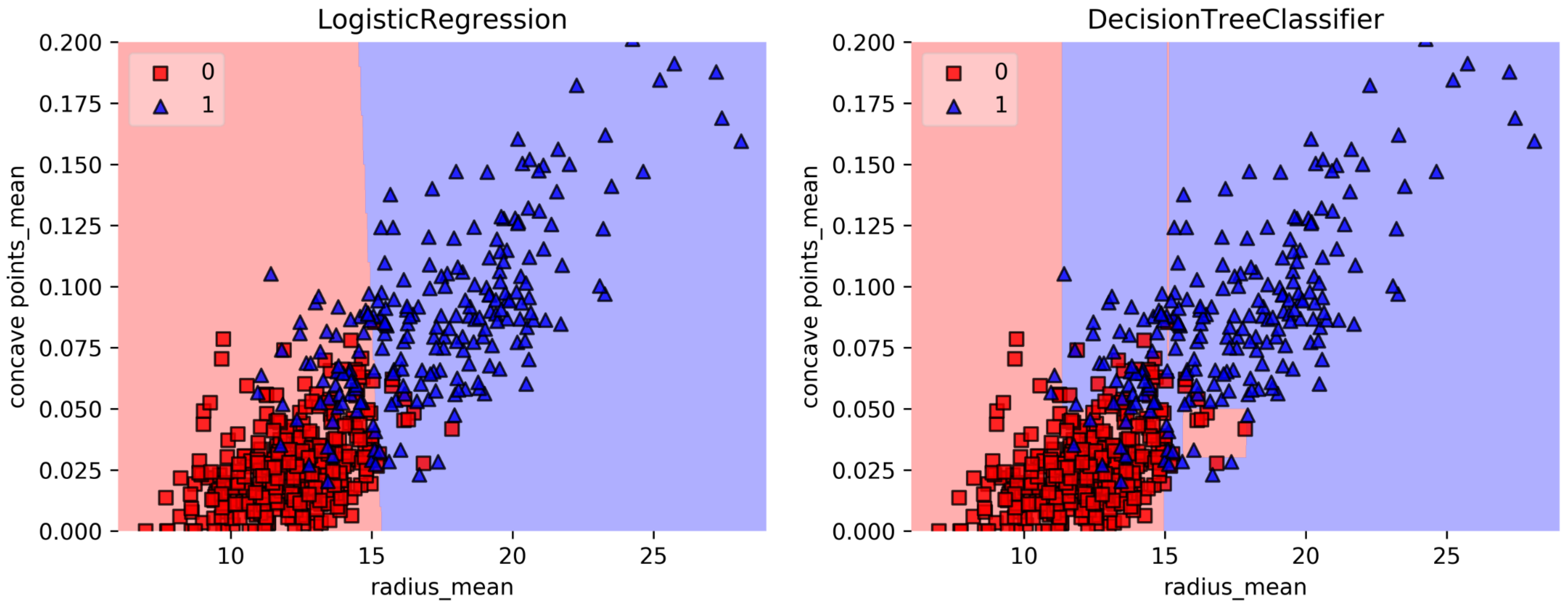
# Decision Regions

**Decision region:** region in the feature space where all instances are assigned to one class label.

**Decision Boundary:** surface separating different decision regions.



# Decision Regions: CART vs. Linear Model



# Let's practice!

# Classification-Tree Learning

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# Building Blocks of a Decision-Tree

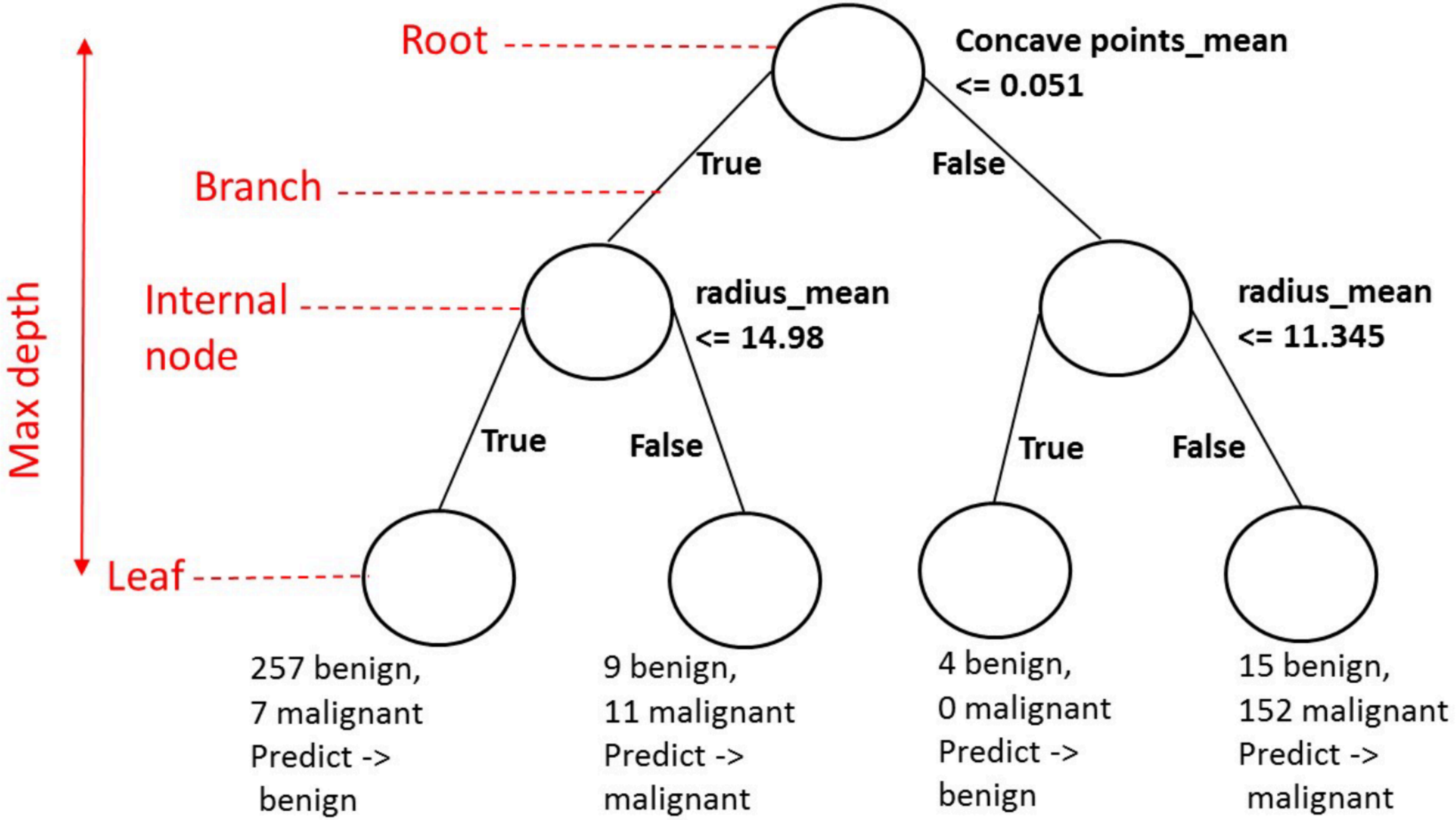
- **Decision-Tree:** data structure consisting of a hierarchy of nodes.
- **Node:** question or prediction.

# Building Blocks of a Decision-Tree

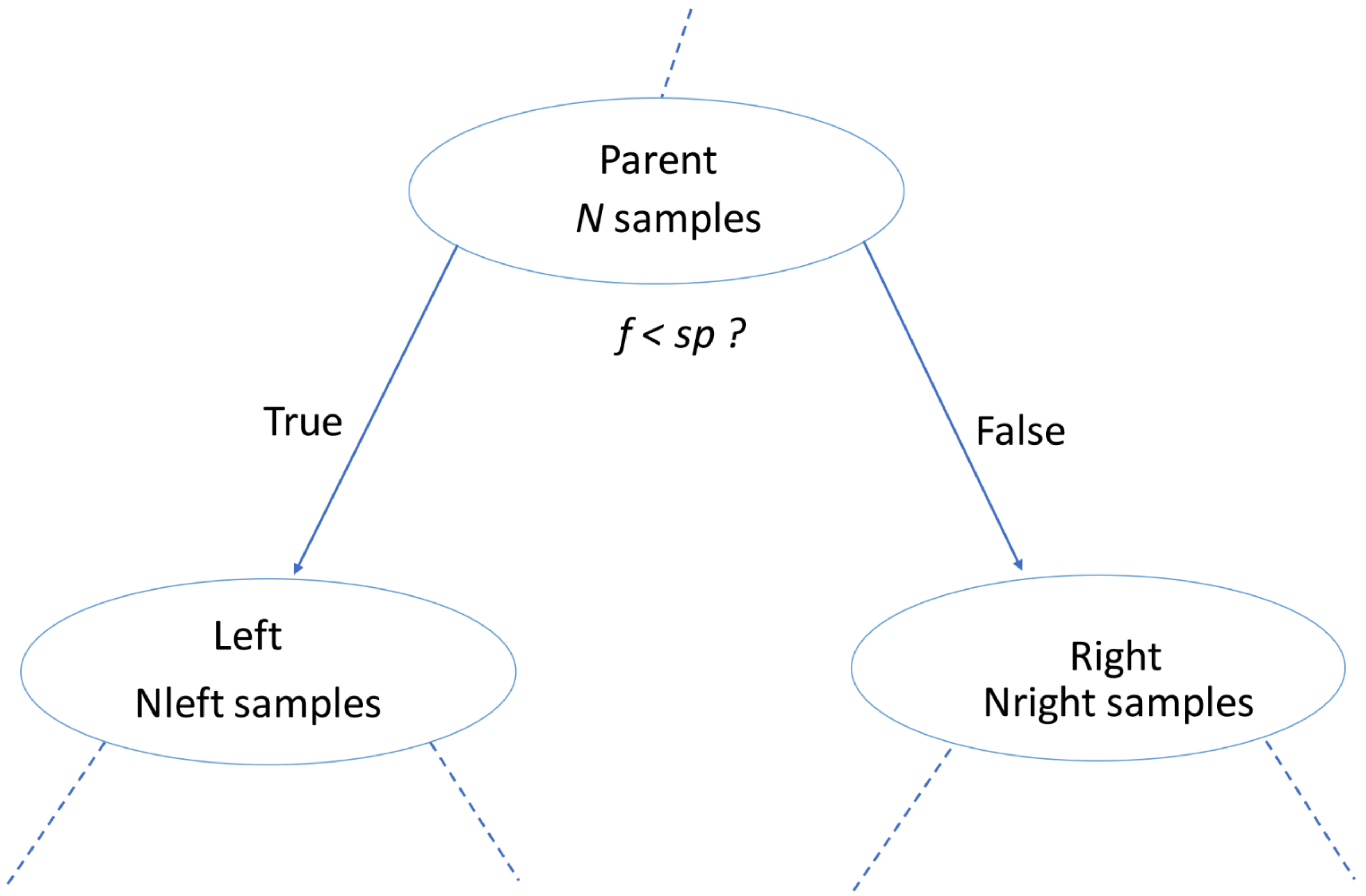
Three kinds of nodes:

- **Root**: *no* parent node, question giving rise to *two* children nodes.
- **Internal node**: *one* parent node, question giving rise to *two* children nodes.
- **Leaf**: *one* parent node, *no* children nodes --> *prediction*.

# Prediction


Root
Branch
Internal node
Leaf
Max depth
Concave points_mean
<= 0.051
True
False
radius_mean
<= 14.98
radius_mean
<= 11.345
True
False
True
False
257 benign,
7 malignant
Predict ->
benign
9 benign,
11 malignant
Predict ->
malignant
4 benign,
0 malignant
Predict ->
benign
15 benign,
152 malignant
Predict ->
malignant

# Information Gain (IG)



# Information Gain (IG)

$$IG(\underbrace{f}_{feature}, \underbrace{sp}_{split-point}) = I(parent) - \left(\frac{N_{left}}{N}\ I(left) + \frac{N_{right}}{N}\ I(right)\right)$$

Criteria to measure the impurity of a node $I(node)$:

- gini index,
- entropy. ...

# Classification-Tree Learning

- Nodes are grown recursively.
- At each node, split the data based on:
  - feature $f$ and split-point $sp$ to maximize $IG(\mathrm{node})$.
- If $IG(\mathrm{node})$= 0, declare the node a leaf. ...

```python
# Import DecisionTreeClassifier
from sklearn.tree import DecisionTreeClassifier
# Import train_test_split
from sklearn.model_selection import train_test_split
# Import accuracy_score
from sklearn.metrics import accuracy_score
# Split dataset into 80% train, 20% test
X_train, X_test, y_train, y_test= train_test_split(X, y,
                                                    test_size=0.2,
                                                    stratify=y,
                                                    random_state=1)
# Instantiate dt, set 'criterion' to 'gini'
dt = DecisionTreeClassifier(criterion='gini', random_state=1)
```

# Information Criterion in scikit-learn

```
# Fit dt to the training set
dt.fit(X_train,y_train)

# Predict test-set labels
y_pred= dt.predict(X_test)

# Evaluate test-set accuracy
accuracy_score(y_test, y_pred)
```

```
0.92105263157894735
```

# Let's practice!

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# Decision-Tree for Regression

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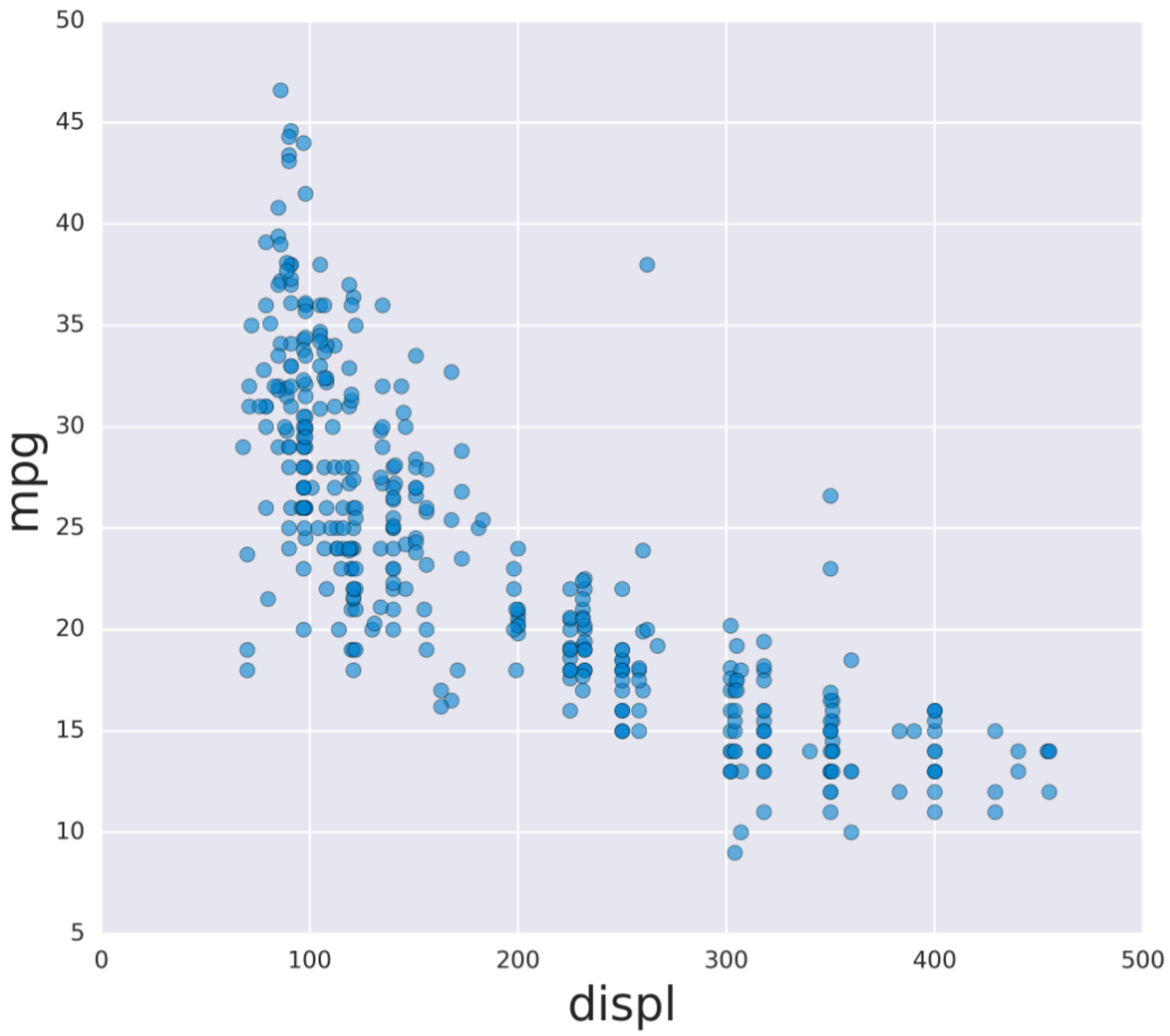
# Auto-mpg Dataset

| | mpg | displ | hp | weight | accel | origin | size |
|---|---|---|---|---|---|---|---|
| **0** | 18.0 | 250.0 | 88 | 3139 | 14.5 | US | 15.0 |
| **1** | 9.0 | 304.0 | 193 | 4732 | 18.5 | US | 20.0 |
| **2** | 36.1 | 91.0 | 60 | 1800 | 16.4 | Asia | 10.0 |
| **3** | 18.5 | 250.0 | 98 | 3525 | 19.0 | US | 15.0 |
| **4** | 34.3 | 97.0 | 78 | 2188 | 15.8 | Europe | 10.0 |
| **5** | 32.9 | 119.0 | 100 | 2615 | 14.8 | Asia | 10.0 |

# Auto-mpg with one feature



# Regression-Tree in scikit-learn

```
# Import DecisionTreeRegressor
from sklearn.tree import DecisionTreeRegressor
# Import train_test_split
from sklearn.model_selection import train_test_split
# Import mean_squared_error as MSE
from sklearn.metrics import mean_squared_error as MSE
# Split data into 80% train and 20% test
X_train, X_test, y_train, y_test= train_test_split(X, y,
                                                   test_size=0.2,
                                                   random_state=3)
# Instantiate a DecisionTreeRegressor 'dt'
dt = DecisionTreeRegressor(max_depth=4,
                           min_samples_leaf=0.1,
                           random_state=3)
```

# Regression-Tree in scikit-learn

```
# Fit 'dt' to the training-set
dt.fit(X_train, y_train)
# Predict test-set labels
y_pred = dt.predict(X_test)
# Compute test-set MSE
mse_dt =  MSE(y_test, y_pred)
# Compute test-set RMSE
rmse_dt = mse_dt**(1/2)
# Print rmse_dt
print(rmse_dt)
```

```
5.1023068889
```

# Information Criterion for Regression-Tree

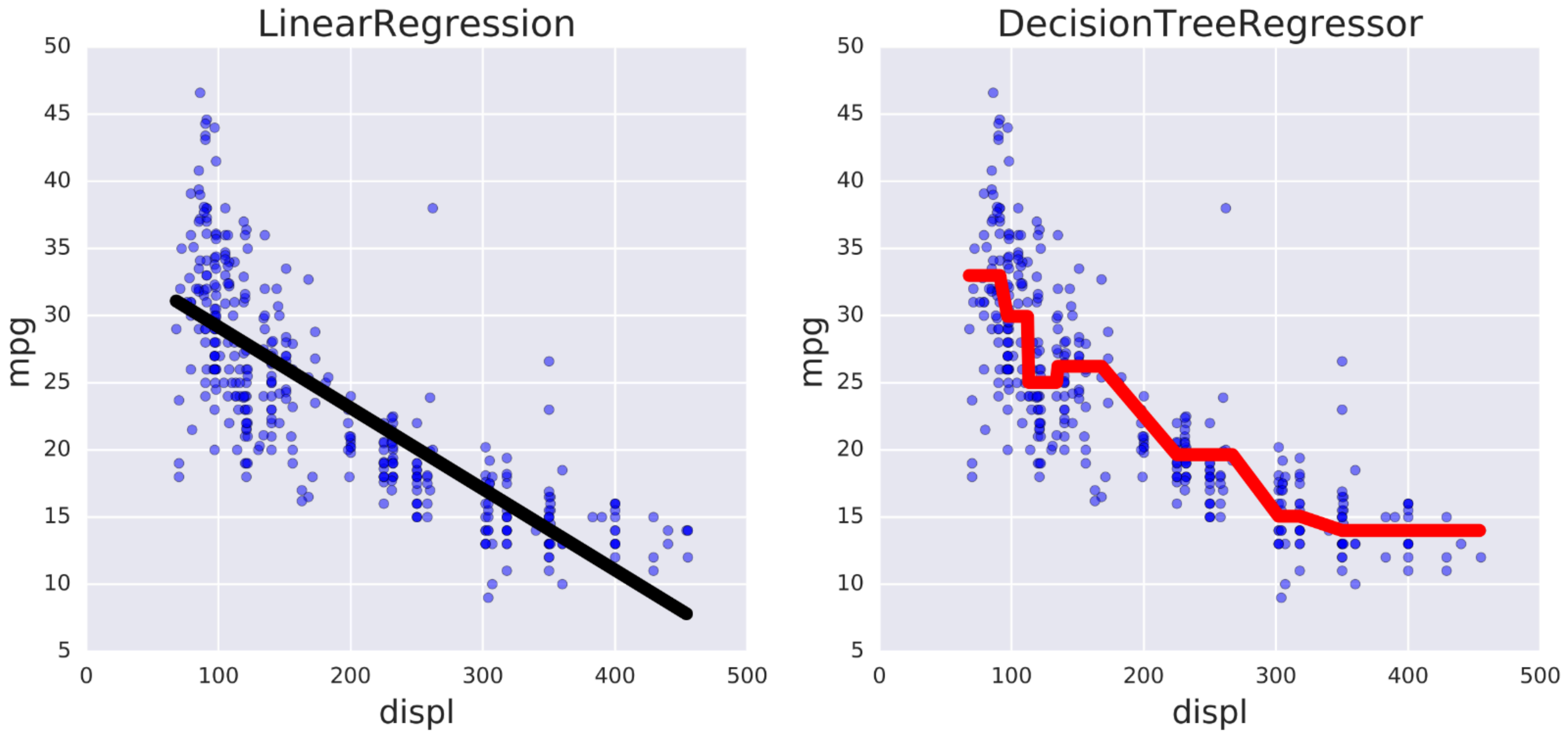
$$I(\text{node}) = \underbrace{\text{MSE}(\text{node})}_{mean-squared-error} = \frac{1}{N_{node}} \sum_{i \in node} \left(y^{(i)} - \hat{y}_{node}\right)^2$$

$$\underbrace{\hat{y}_{node}}_{mean-target-value} = \frac{1}{N_{node}} \sum_{i \in node} y^{(i)}$$

# Prediction

$$\hat{y}_{pred}(\text{leaf}) = \frac{1}{N_{leaf}} \sum_{i \in leaf} y^{(i)}$$

# Linear Regression vs. Regression-Tree



# Let's practice!

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